

# Rohith Reddy G K

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🔗 [RohithReddyGK](https://github.com/RohithReddyGK)

🌐 [rohith-portfolio-gk](https://github.com/rohith-portfolio-gk)

## Skills

**Languages:** Java, Python, JavaScript, SQL, NoSQL

**Frameworks & Libraries:** React.js, Tailwind CSS, Flask, Django, Chart.js, Recharts

**Databases:** Oracle/SQL, MongoDB, PostgreSQL

**Tools & Platforms:** Git, GitHub, Power BI, Tableau, Docker, Version Control (Git/GitHub)

**Core Competencies:** DSA, OOPS, REST APIs, ML Model Training & Deployment, CI/CD, Cloud Deployment

## Projects

### ADMET-X: AI-Driven Platform for In-silico ADMET Prediction | [GitHub](#)

*Tech Stack: React.js, Tailwind CSS, Flask, RDKit, RESTful APIs*

- Developed a full-stack AI platform for in-silico ADMET prediction using Flask REST APIs and React.js frontend, improving drug candidate screening efficiency by **45%**.
- Processed **31** drug datasets (~**100K** compounds) and generated molecular descriptors using RDKit to enhance data quality.
- Trained and validated Random Forest and XGBoost models achieving **90%+** accuracy and **85%+** R<sup>2</sup> score.
- Deployed on cloud environment enabling real-time predictions with seamless UI integration.

### AI-Driven Code Reviewer | [GitHub](#)

*Tech Stack: React, Vite, Tailwind CSS, Flask, Python, Groq LLM API, Monaco Editor*

- Built an AI-powered code review platform that analyzes source code, detects bugs and anti-patterns, and generates optimized code alternatives using Groq's low-latency LLM inference.
- Integrated a Monaco-based in-browser code editor with syntax highlighting and automatic language-mismatch detection across 6 programming languages.
- Implemented time and space complexity analysis with visual charts, one-click "Apply" for AI-optimized code, and a context-aware AI chat assistant.
- Added timestamped analysis history and one-click PDF report export; deployed the full-stack app (React frontend, Flask backend) on Vercel.

### AI Powered Dashboard & Data Chatbot | [GitHub](#)

*Tech Stack: Next.js, TailwindCSS, Recharts, Express.js, PostgreSQL, Groq LLM, Vanna AI*

- Built an AI-enabled financial analytics dashboard with natural language query support for real-time insights.
- Integrated LLM-based query understanding (Vanna + Groq) to dynamically generate and execute SQL queries via Express APIs.
- Designed interactive Recharts visualizations, real-time metrics, and NLP summarization features reducing manual analysis time by **40%**.
- Implemented secure backend APIs ensuring **100%** response reliability for large dataset operations.

## Experience

### Healthcare Technology Innovation Centre (HTIC), IIT Madras

Chennai, India

*Project Associate – Image-Guided Robotic Surgery (IGRS) Team*

*Jun 2026 – Present*

- Developing deep learning models for 3D reconstruction of anatomical structures from medical imaging data to support surgical navigation.
- Designing, training, and optimizing AI/ML deep learning models to solve real-time clinical problems in image-guided robotic surgery and diagnostic imaging.

*Project Intern – Image-Guided Robotic Surgery (IGRS) Team*

*Feb 2026 – May 2026*

- Worked with the IGRS team on research-oriented medical technology systems, gaining hands-on exposure to clinical-grade medical robotics and imaging systems.

### AI Internship

Remote

*Infosys Springboard*

*Feb 2026 – Apr 2026*

- Designed and developed an intelligent "AI-Driven Code Reviewer" system as part of the Infosys Springboard AI Virtual Internship, analyzing source code and generating automated review suggestions using AI-based techniques.
- Progressed through 4 structured milestones covering project architecture, AI concepts, and implementation strategies; independently carried out complete project design, development, and integration.

## Education

### AMC Engineering College, Bengaluru

Bengaluru, India

*B.E., Information Science & Engineering | CGPA: 9.22*

*Dec 2022 – Jul 2026*

## Certifications & Achievements

### IEEE Publication:

"ADMET-X: Machine Learning Framework for Predicting Pharmacokinetic and Toxicological Drug Properties"

### IJCRT Publication: "IoT-Based Automatic Medicine Reminder"

**NPTEL:** Data Structures & Algorithms (Java), and Programming in Java.

**Microsoft Certified:** Generative AI – LinkedIn Learning.